EXP-6

Use Figma to simulate a handoff scenario in a mobile network and visualize the process

1. Aim

To design and develop an interactive prototype using Figma to simulate the handoff process in a mobile network and visualize how an ongoing mobile connection is transferred from one base station to another without interruption when a user moves between different cell areas.

2. Software Used

- Software: Figma
- Platform: Web-based design tool
- Application: UI Design and Prototyping
- Internet: Required to access Figma online.

3. Algorithm

- Start the Figma application and create a new design file.
- Create mobile screen frames for the handoff simulation.
- Design the welcome screen explaining mobile network handoff.
- Display the concept of handoff and its importance
- Illustrate the mobile network setup with two base stations (BS1 and BS2).
- Show the movement of a mobile user from the coverage area of BS1 towards BS2.
- Compare the signal strengths received from both base stations.
- Check whether the signal from BS2 is sufficiently stronger and suitable for handoff.

- If YES, transfer the mobile connection from BS1 to BS2.
- If NO, continue monitoring the signal while maintaining the existing connection.
- Create navigation links between the screens using the Prototype feature.
- Run the prototype and verify the handoff simulation.
- Stop.

4. Design

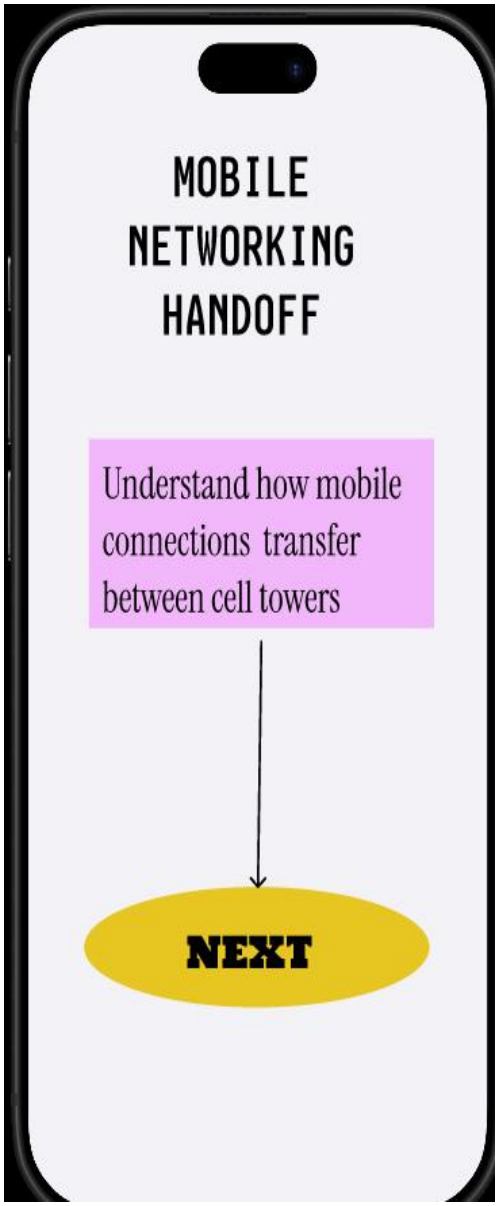
The handoff simulation is designed using seven mobile screen frames.

Screen	**Design description**
1. Welcome	Introduction to mobile network handoff with a Start button.
2. What is Handoff?	Explains the meaning and importance of handoff.
3. Network Setup	Shows mobile station, BS1 and BS2.
4. Mobile User Movement	Illustrates movement from BS1 coverage towards BS2.
5. Handoff Decision	Compares signal strengths and displays the handoff decision.
6. Successful Handoff	Shows connection transfer from BS1 to BS2.
7. Thank You	Displays the completion message.

Design elements used:

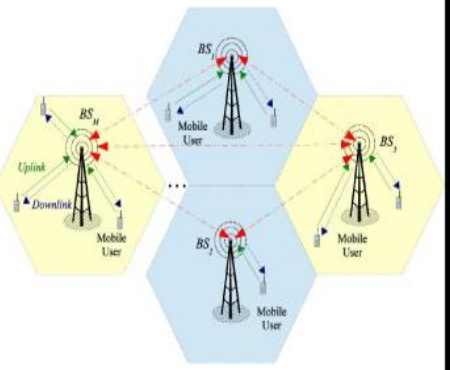
- Mobile phone frames to represent application screens.
- Shapes and circles to represent base stations and coverage areas.

- Arrows to indicate the direction of user movement and connection transfer.
- Text labels to explain each stage.
- Colours to differentiate BS1 and BS2.
- Buttons for navigation between screens.





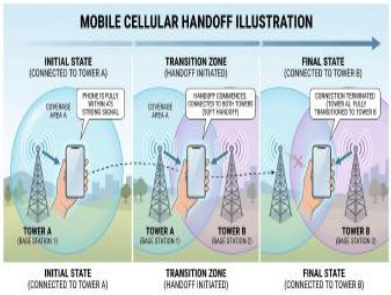
MOBILE NETWORK SETUP



NETWORK COMPONENTS

Mobile Station (MS): Device used by the mobile user to communicate.
Base Station 1 (BS1): Initially serves the mobile user.
Base Station 2 (BS2): Takes over when the user moves into its coverage area.

Mobile User Movement

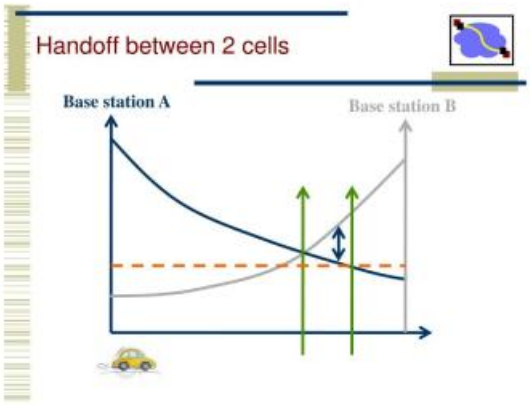


USER MOVEMENT PROCESS

The mobile user is initially connected to BS1.
The user moves away from BS1 coverage.
The signal strength from BS1 gradually decreases.
BS2 becomes the next suitable base station.

iPhone 18 Pro - 5

HANDOFF DECISION



NETWORK CHECKS SIGNAL STRENGTH

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IS BS2 SIGNAL STRONGER AND SUITABLE?

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YES

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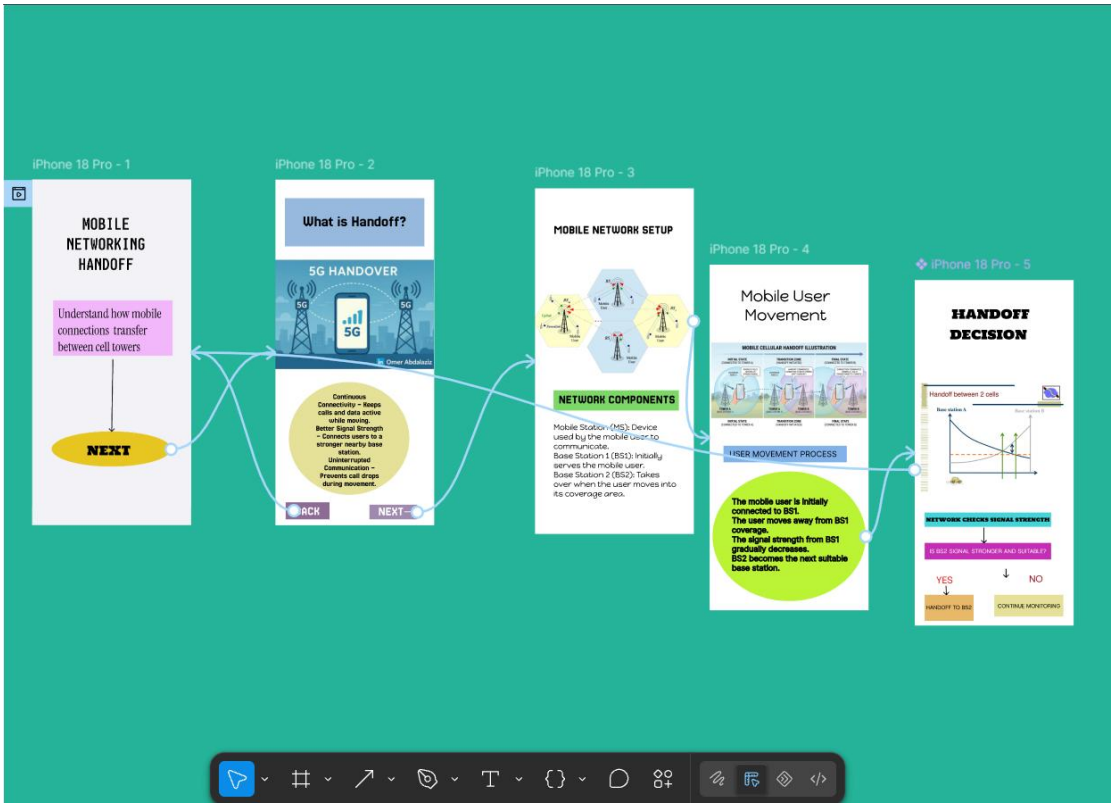
HANDOFF TO BS2

NO

↓

CONTINUE MONITORING

5. Prototype



Result:

Thus, the mobile network handoff process was successfully designed and simulated using Figma. An interactive prototype was created to visualize mobile user movement, signal strength comparison, handoff decision-making, and successful transfer of connection between two base stations.